Linux Mentorship Program - Cover Letter

Hi,

My name is K Hasini Reddy and I'm a Computer Science student at IIIT Vadodara, India. I am writing to apply for the Linux Mentorship Program for the project "Mapping the RISC-V Extensions Landscape".

How I found out about the program

I came across this program while I was browsing through open source opportunities that matched my current skills and current interests. This specific project stood out immediately because it combines two things I already work with; React and data engineering, applied to something I want to learn deeply: the RISC-V ISA.

Why I am interested

This project sits exactly at the intersection I find most interesting. On one side there is low-level ISA architecture; ratified extensions, encoding diagrams and on the other side there is a well-structured React interface that makes that complexity accessible. What drew me in more was the specifics of the problems: 137 extensions missing instruction data, 104 mismatched tags, sync tooling gaps. These tasks are concrete and meaningful. Contributing to a tool which is already live and is used by real people exploring RISC-V, is exactly the kind of work I want to do.

What I bring to the program

The three tasks described maps closely to things I've already done:

- For the data reconciliation task: matching 104 mismatched extension tags from riscv-opcodes to the catalog; I have experience working with structured data pipelines and naming inconsistencies from my open-source work at CircuitVerse. I am comfortable writing scripts to detect, report, and resolve mismatches systematically.
- For the instruction mapping task: researching and filling in supervisor-level and vector subset extensions; I have done this kind of ground-up research before. I learned Solidity from scratch for my Smart Fisheries project because the problem needed it. RISC-V is new to me, but structured and spec-driven research is not.

- For the tooling improvement task: building gap detection and coverage reports; this is close to work I have done in automated validation and reporting in my projects. I understand what good developer tooling looks like and how to make sync scripts reliable.

In the frontend, the project uses React and Tailwind CSS which are already part of my regular workflow.

What I hope to get out of this experience

I want to come out of this with two things. First, a real understanding of the RISC-V ISA, not surface-level, but deep enough to contribute accurately to a reference tool that developers rely on. Second, experience working on an open-source project at a scale and standard that my personal projects cannot ofcourse replicate. I want to learn how a maintainer like this project's mentor thinks about data quality, coverage, and tooling design, and carry that forward into everything I build after this.

Thank you!

K Hasini Reddy.